

I. THEORETICAL PROPERTIES OF WCS MATCHING

We first establish that the WCS objective $H_0(\mathbf{X})$ is invariant under a joint relabeling of the vertices of G and H , and then use this to characterize when *two distinct* correspondence matrices for the *same* pair of graphs G, H are nonetheless equally optimal.

Proposition 1 (Relabeling Invariance). *Let $\mathbf{P} \in \{0, 1\}^{M \times M}$ and $\mathbf{Q} \in \{0, 1\}^{N \times N}$ be permutation matrices, and let*

$$\mathbf{A}'_G = \mathbf{P}\mathbf{A}_G\mathbf{P}^T, \quad \mathbf{A}'_H = \mathbf{Q}\mathbf{A}_H\mathbf{Q}^T \quad (1)$$

denote G and H under a relabeling of their vertices. Then, for any $\mathbf{X} \in \mathcal{P}$,

$$H_0(\mathbf{P}\mathbf{X}\mathbf{Q}^T; \mathbf{A}'_G, \mathbf{A}'_H) = H_0(\mathbf{X}; \mathbf{A}_G, \mathbf{A}_H). \quad (2)$$

Consequently, if $\mathbf{X}^$ minimizes $H_0(\cdot; \mathbf{A}_G, \mathbf{A}_H)$ over $\mathcal{P}$, then $\mathbf{P}\mathbf{X}^*\mathbf{Q}^T$ minimizes $H_0(\cdot; \mathbf{A}'_G, \mathbf{A}'_H)$ over $\mathcal{P}$, and the two problems attain the same optimal value.*

Proof. Let $\mathbf{X}' = \mathbf{P}\mathbf{X}\mathbf{Q}^T$. Since $\mathbf{Q}$ is a permutation matrix, $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}_N$ and $\mathbf{Q}^T\mathbf{1}_{N \times N}\mathbf{Q} = \mathbf{1}_{N \times N}$, so

$$\mathbf{U}' = \mathbf{X}'\mathbf{1}_{N \times N}\mathbf{X}'^T = \mathbf{P}(\mathbf{X}\mathbf{1}_{N \times N}\mathbf{X}^T)\mathbf{P}^T = \mathbf{P}\mathbf{U}\mathbf{P}^T. \quad (3)$$

Since conjugation by a permutation matrix commutes with the Hadamard product, $(\mathbf{P}\mathbf{A}\mathbf{P}^T) \circ (\mathbf{P}\mathbf{B}\mathbf{P}^T) = \mathbf{P}(\mathbf{A} \circ \mathbf{B})\mathbf{P}^T$, we obtain

$$\mathbf{U}' \circ \mathbf{A}'_G = \mathbf{P}(\mathbf{U} \circ \mathbf{A}_G)\mathbf{P}^T, \quad \mathbf{X}'\mathbf{A}'_H\mathbf{X}'^T = \mathbf{P}(\mathbf{X}\mathbf{A}_H\mathbf{X}^T)\mathbf{P}^T, \quad (4)$$

where the second identity uses $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}_N$. Subtracting the two expressions in (4) gives

$$(\mathbf{U}' \circ \mathbf{A}'_G) - \mathbf{X}'\mathbf{A}'_H\mathbf{X}'^T = \mathbf{P}[(\mathbf{U} \circ \mathbf{A}_G) - \mathbf{X}\mathbf{A}_H\mathbf{X}^T]\mathbf{P}^T. \quad (5)$$

Because $\mathbf{P}$ is orthogonal ($\mathbf{P}\mathbf{P}^T = \mathbf{I}_M$), the Frobenius norm satisfies $\|\mathbf{P}\mathbf{Z}\mathbf{P}^T\|_F = \|\mathbf{Z}\|_F$ for any square matrix $\mathbf{Z}$; applying this to (5) yields (2). Finally, $\mathbf{X} \mapsto \mathbf{P}\mathbf{X}\mathbf{Q}^T$ is a bijection of $\mathcal{P}$ onto itself, since $\mathbf{P}, \mathbf{Q}$ are invertible and preserve the row-sum, column-sum, and cardinality constraints defining $\mathcal{P}$. Hence minimizing $H_0(\cdot; \mathbf{A}_G, \mathbf{A}_H)$ over $\mathcal{P}$ is equivalent, under this bijection, to minimizing $H_0(\cdot; \mathbf{A}'_G, \mathbf{A}'_H)$ over $\mathcal{P}$, with identical optimal value. $\square$

Proposition 1 concerns *two different pairs of graphs* $(\mathbf{A}_G, \mathbf{A}_H)$ and $(\mathbf{A}'_G, \mathbf{A}'_H)$ related by a relabeling. A natural question is what happens when $\mathbf{P}$ and $\mathbf{Q}$ leave the graphs themselves unchanged, i.e., when they are *automorphisms*. This yields the following corollary, which explains why a WCS instance can admit multiple, structurally distinct, equally optimal correspondence matrices.

Corollary 1 (Non-Uniqueness under Graph Automorphisms). *Let $\mathbf{P} \in \text{Aut}(G)$ and $\mathbf{Q} \in \text{Aut}(H)$ be automorphisms of G and H , respectively, i.e., permutation matrices satisfying*

$$\mathbf{P}\mathbf{A}_G\mathbf{P}^T = \mathbf{A}_G, \quad \mathbf{Q}\mathbf{A}_H\mathbf{Q}^T = \mathbf{A}_H. \quad (6)$$

Then, for any $\mathbf{X} \in \mathcal{P}$,

$$H_0(\mathbf{P}\mathbf{X}\mathbf{Q}^T; \mathbf{A}_G, \mathbf{A}_H) = H_0(\mathbf{X}; \mathbf{A}_G, \mathbf{A}_H). \quad (7)$$

In particular, if $\mathbf{X}^$ is an optimal WCS correspondence for $(\mathbf{A}_G, \mathbf{A}_H)$, then $\mathbf{Y}^* = \mathbf{P}\mathbf{X}^*\mathbf{Q}^T$ is also optimal for the same pair $(\mathbf{A}_G, \mathbf{A}_H)$, generally with $\mathbf{Y}^* \neq \mathbf{X}^*$ whenever $(\mathbf{P}, \mathbf{Q}) \neq (\mathbf{I}_M, \mathbf{I}_N)$. Hence the optimal solution set of a WCS instance is a union of orbits under the action of $\text{Aut}(G) \times \text{Aut}(H)$, and is a singleton only if this action is trivial on the optimal correspondence.*

Proof. Apply Proposition 1 with $\mathbf{A}'_G = \mathbf{P}\mathbf{A}_G\mathbf{P}^T$ and $\mathbf{A}'_H = \mathbf{Q}\mathbf{A}_H\mathbf{Q}^T$. By (6), $\mathbf{A}'_G = \mathbf{A}_G$ and $\mathbf{A}'_H = \mathbf{A}_H$, so (2) reduces directly to (7), with both sides evaluated on the same, unlabeled pair $(\mathbf{A}_G, \mathbf{A}_H)$. $\square$

Remark 1. *Corollary 1 should be distinguished from Proposition 1: the latter compares the same matching problem written in two different coordinate systems, while the former shows that a single WCS instance can have several distinct optimal matrices $\mathbf{X}^*, \mathbf{Y}^* = \mathbf{P}\mathbf{X}^*\mathbf{Q}^T, \dots$ whenever G or H has nontrivial symmetry, all attaining the identical objective value $H_0(\mathbf{X}^*)$. Conversely, two matrices that are not related by an automorphism pair $(\mathbf{P}, \mathbf{Q}) \in \text{Aut}(G) \times \text{Aut}(H)$ generally attain different objective values, so (7) does not hold for an arbitrary pair of feasible correspondences.*